LEARNING AND ASSIGNMENT OF IP ADDRESS MANUALLY TO COMPUTERS.

AIM:

To understand the IP addressing scheme and manually assign static IP addresses to computers on a network.

THEORY:

An IP (Internet Protocol) address is a unique numerical label assigned to each device connected to a computer netwo
IPV4 addresses are 32 bit number written in dotted-decim notation.

Subnet mask determines the network and host portions.

Manual (static) IP assignment is used for servers and network devices requiring a permanent address DHCP assigns IPs dynamically.

PROCEDURE:

In Windows: Open Network and sharing center → change ad settings → Right click.

Local area connection → properties → IPV4 → set IP man

* Enter IP address: 192.168.1.10, Subnet mask: 255.255.255
Default gateway: 192.168.1.1.
* Enter DNS server: 8.8.8.8 (Google DNS), click OK.
* In Linux: Use command 'sudo ip 'addr add 192.168.1.
dev eth0',

* To get gateway in linux = 'sudo ip route add default via 192.168.1.1'.
* For persistent configuration in Linux, edit /etc/network/interfaces or use nmcli.
* Verify the configuration using 'ipconfig' or 'ipa'.
* Ping the default gateway to confirm network connectivity.
* Ping another computer on the same subnet to verify communication.

RESULT:

Static IP addresses were successfully assigned to con
Connectivity was verified using ping commands between
hosts on the same subnet.